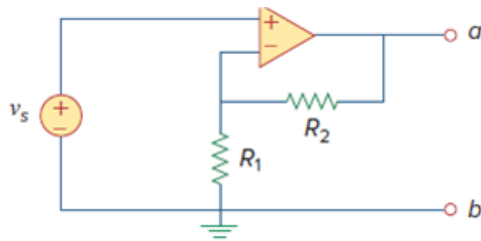


Problem

For the circuit shown in Fig. 5.73, find the Theveninequivalent at terminals a - b . (*Hint: To find R_{Th} , apply a current source i_o and calculate v_o .*)

Figure. 5.73:



Step-by-step solution

Step 1 of 3

$V_{TH} = V_{ab}$.

But,

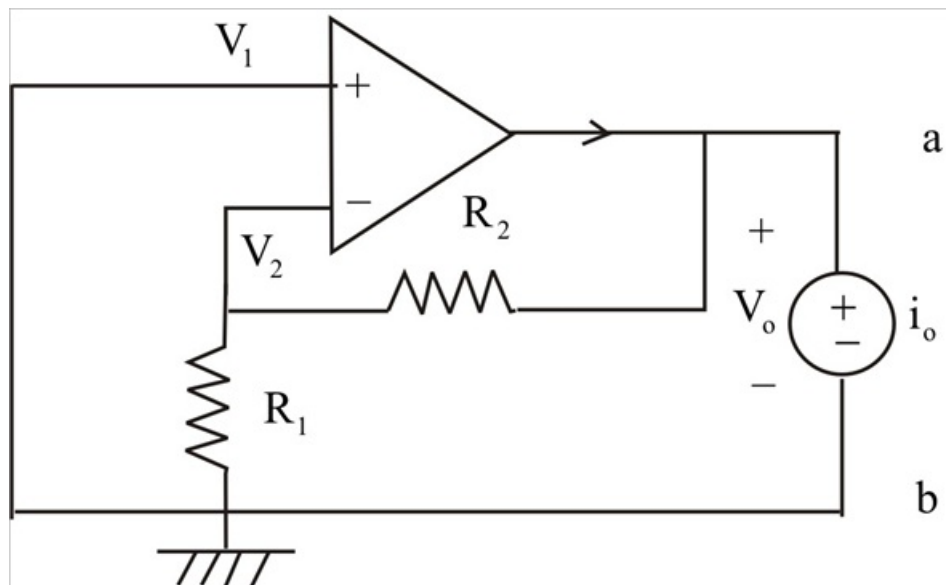
$$V_s = \frac{R_1}{R_1 + R_2} V_{ab} \quad .$$

Thus,

$$V_{TH} = V_{ab} = \frac{R_1 + R_2}{R_1} V_s = \left(1 + \frac{R_2}{R_1} \right) V_s \quad .$$

To get R_{TH} , Apply a current source i_o at terminals $a - b$ as shown below.

Step 2 of 3



Step 3 of 3

Since the non-inverting terminal is connected to ground, $V_1 = V_2 = 0$.

i.e., No current passes through R1 and consequently R2.

Thus,

$$V_0 = 0 \quad \text{And} \quad R_{TH} = \frac{V_0}{i_0} = 0 .$$